

YUVA Cybersecurity Internship

Comprehensive 4-Week Technical Report: Infrastructure Assessment, Threat Modeling,
Incident Response & Strategic Defense

Author: Deepika Jaiswal | **Student ID:** 2410030120

Role: Junior Cyber Security Analyst

Target System: AuraGov Citizen Services Portal (auragov.service.gov)

Executive Summary

Internship Focus

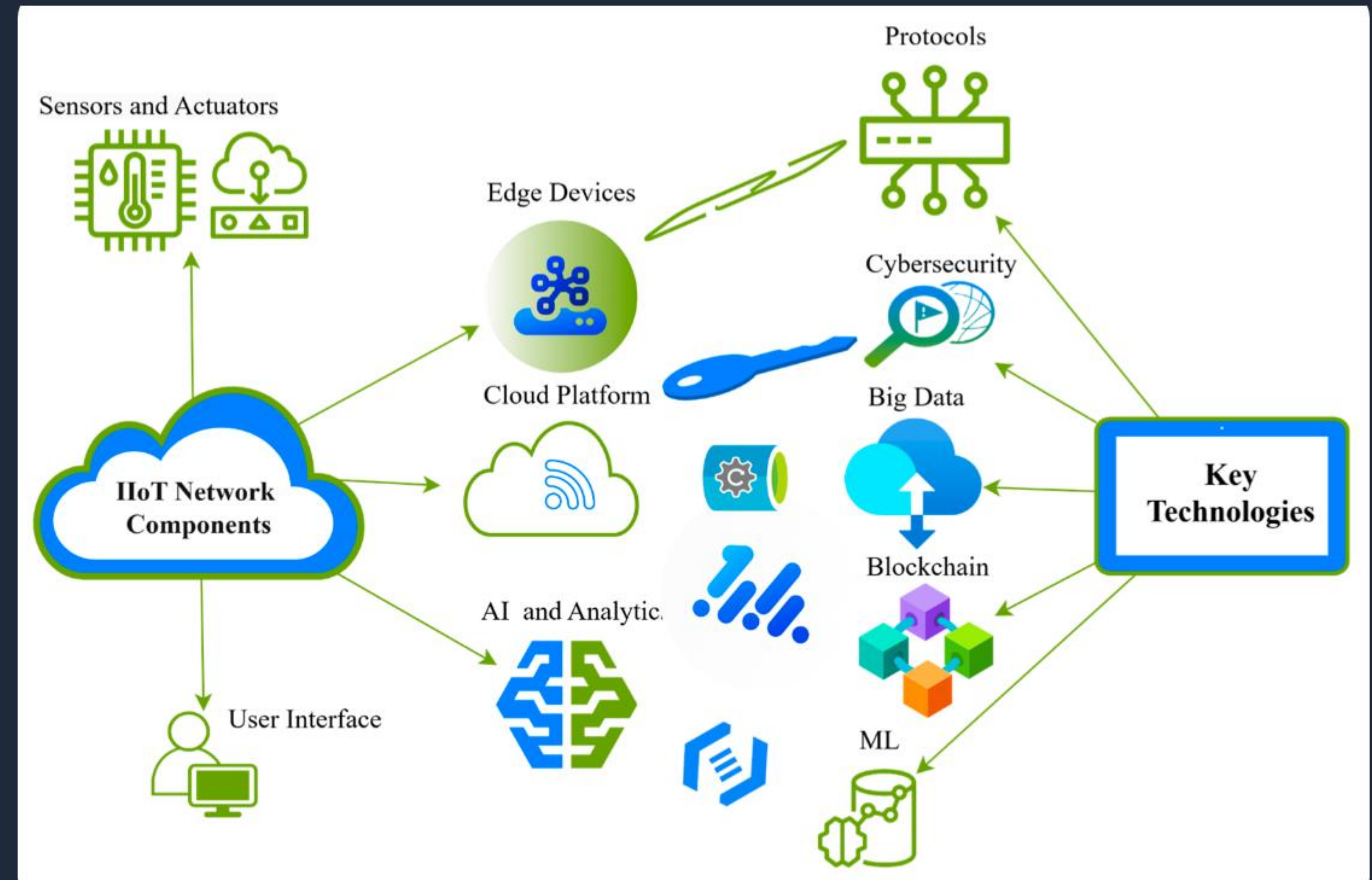
Evaluated the AuraGov National Citizen Services Portal, a cloud-native platform providing over 150 digital public services, including identity verification, property registries, and welfare benefit disbursements.

Key Deliverables

Mapped infrastructure & vulnerabilities (Week 1), analyzed threat vectors & risk priorities (Week 2), executed a simulated incident response (Week 3), and formulated a 4-year Post-Quantum & Zero Trust roadmap (Week 4).

System Architecture

- **Presentation Layer:** Single-Page Applications (React) served globally via Content Delivery Networks.
- **Edge Protection:** Cloudflare WAF terminating public TLS 1.3 and absorbing DDoS attacks.
- **Application Tier:** Spring Boot microservices on Kubernetes clusters interacting with state registries.
- **Data Tier:** Encrypted PostgreSQL database cluster enforcing AES-256 storage encryption and PII tokenization.



Key Vulnerabilities

ID	Vulnerability	Severity	Target Component	Impact Summary
VULN-01	Broken Object Level Authorization	CRITICAL	Property Registry API	Parameter tampering allowed downloading land titles of other citizens.
VULN-02	Weak Token Validation	HIGH	e-Sign Gateway	JWT tokens lacked issuer verification, risking signature forgery.
VULN-03	Insufficient Rate Limiting	MEDIUM	Citizen Login Portal	Missing IP/device rate controls exposed endpoints to brute force.
VULN-04	Missing HTTP Headers	LOW	Web Frontends	Absence of strict CSP and frame options increased clickjacking risk.

Threat Actor Profiles



APTs

Motivated by espionage & disruption.
Target identity registries and database clusters using zero-days and living-off-the-land tools.



Cybercrime Groups

Target financial gain, extortion, and ransomware across payment gateways and welfare disbursement APIs.



Hacktivists & Insiders

Aim to disrupt public portals or abuse administrative access to leak internal data logs and databases.

Risk Model & Priorities

25

Max Risk Score ($R = L \times I$)

MITRE ATT&CK Risk Evaluation

TR-01 (Score 25): BOLA / Data Exfiltration (T1190) - Critical

TR-02 (Score 20): JWT Middleware Token Forgery (T1556) - Critical

TR-03 (Score 16): Kubernetes Internal Lateral Movement (T1046) - High

TR-04 (Score 15): Automated Login Credential Stuffing (T1110) - High

Incident Timeline



Incident Metrics

7m

Mean Time to Detect

SIEM alert triggered within 180 seconds of attack start.

15m

Mean Time to Contain

WAF context block & JWT key revocation executed rapidly.

0m

Service Downtime

Rolling Kubernetes updates ensured seamless user access.

1,240

Exfiltration Cap

Data loss restricted to non-sensitive property titles.

Future Threat Vectors

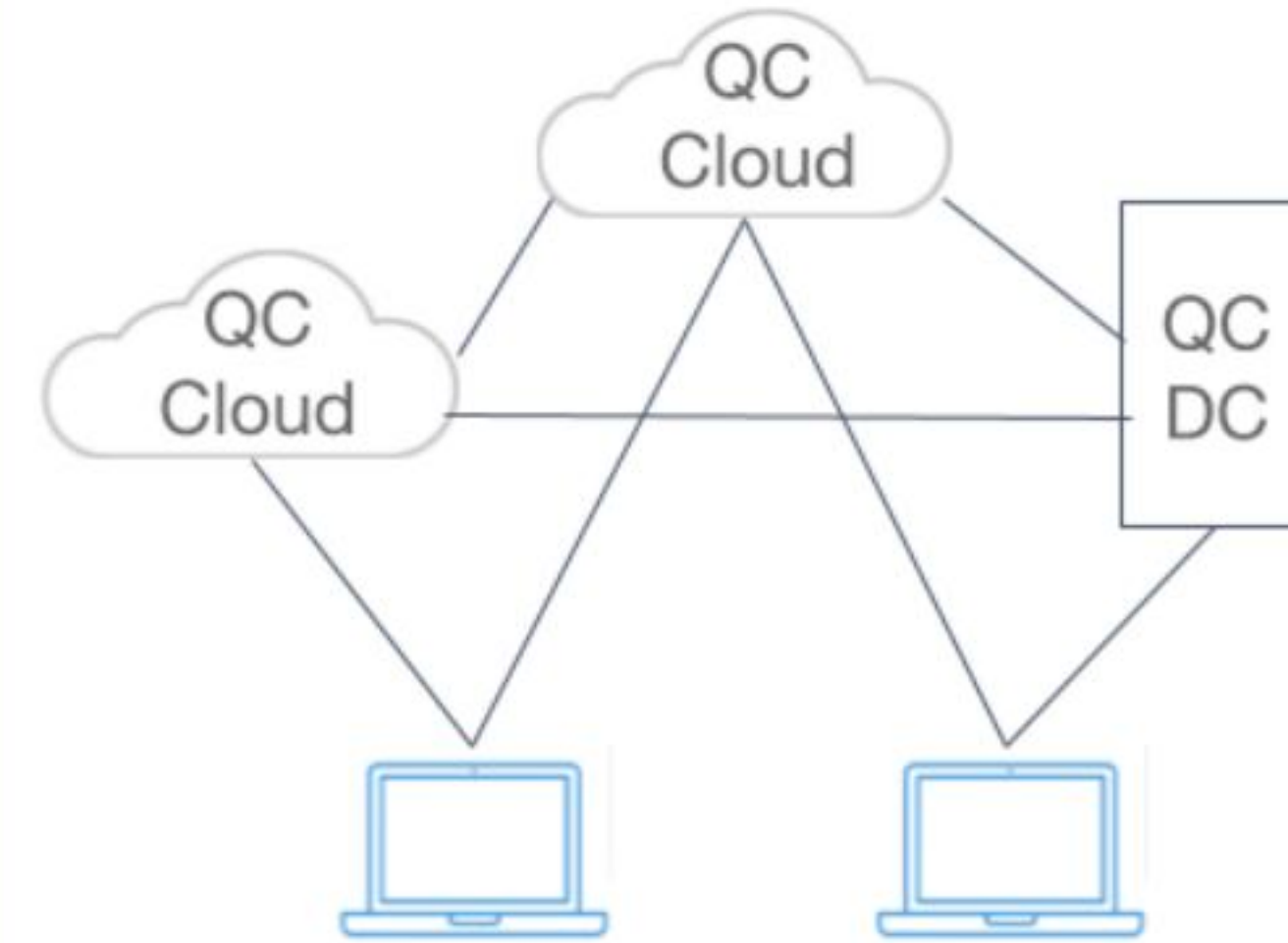
2026–2030 Horizon

- **AI Logic Swarms:** Autonomous agentic AI models probing API logic gaps faster than signature-based WAFs can adapt.
- **Cryptographic Decay:** "Harvest Now, Decrypt Later" risks targeting classical RSA/ECC public key infrastructure.
- **Self-Sovereign Identity:** Migrating to W3C Verifiable Credentials and Zero-Knowledge Proofs (ZKP).

Quantum Communication (QSC)

Distributed & Blind Quantum Compute

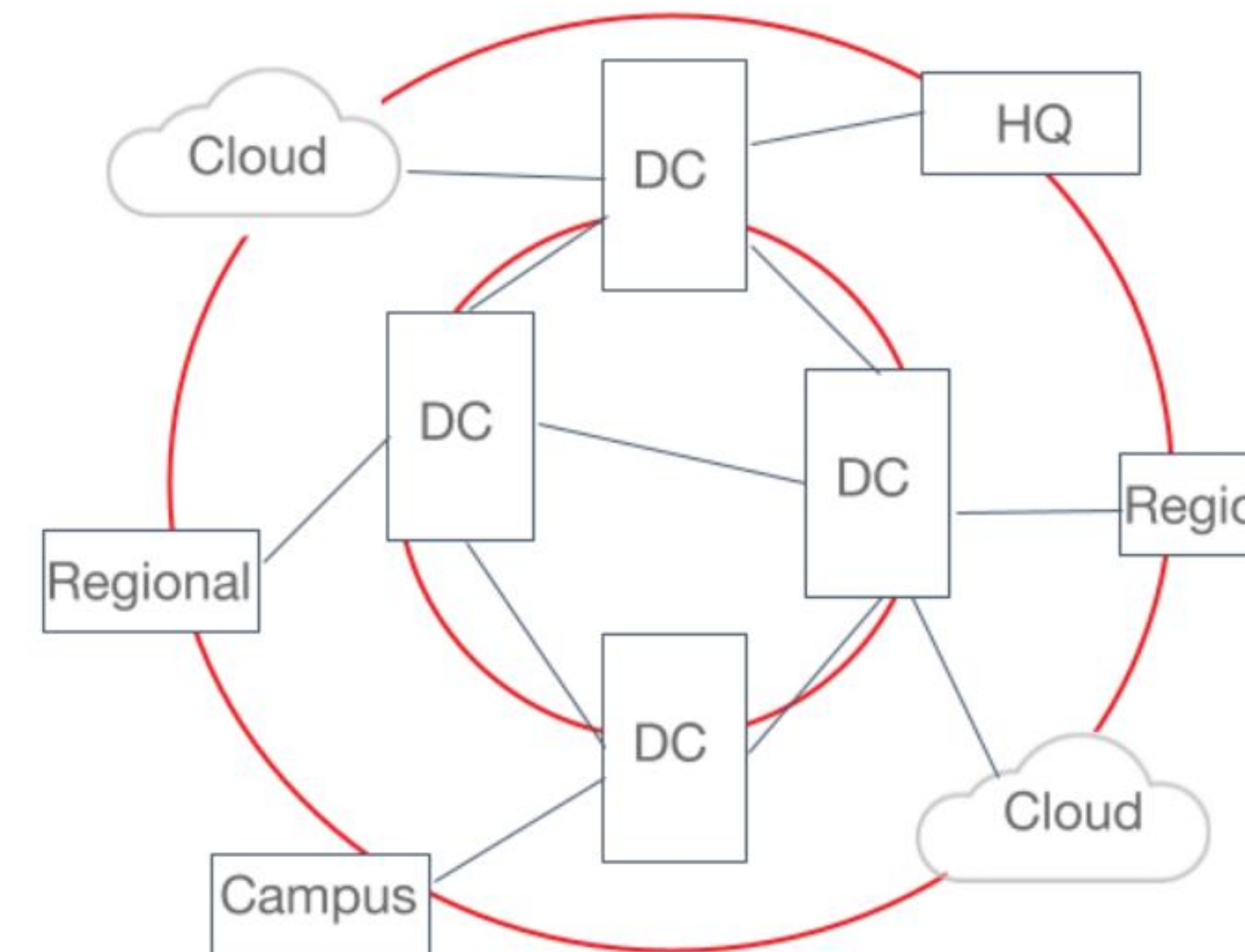
MAN → WAN



Quantum Compute
Scale, Extended Reach,
Cost Reduction,
Generative AI, Privacy
for LLMs, Data, and
Algorithms

Quantum Secure Communications

MAN → WAN



Secure Sensitive Data,
Eavesdropping
Protection, Harvest
Now Decrypt Later
Mitigation, Quantum
App Support

Future Readiness Index

Current Baseline (2026)

FRI = 0.007

Strategic Target (2030)

FRI = 0.518

Quantitative Formula: $FRI = \frac{\sigma_{PQC} \times \alpha_{ZT}}{\delta_{AI} \times \tau_{Legacy}}$ / *Target Goal:* $FRI > 0.500$

Strategic Implementation



Phase 1 (Months 0–12): Deploy AI-driven WAF behavior engines, enforce automated PII log masking filters, and complete a full Post-Quantum Cryptography asset audit.



Phase 2 (Months 12–24): Roll out Istio service mesh mTLS across all Kubernetes pods and deploy hybrid ML-KEM / ML-DSA tokens on e-Sign gateways.



Phase 3 (Months 24–48): Integrate Self-Sovereign Identity (SSI) with Zero-Knowledge Proofs (ZKP) and transition SOC playbooks to autonomous agentic SOAR.

Conclusion & Reflection

Bridging academic theory with practical cybersecurity operations to protect national digital public infrastructure.

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Image Sources



https://www.mdpi.com/sensors/sensors-25-00213/article_deploy/html/images/sensors-25-00213-g004.png

Source: www.mdpi.com



https://www.aliroquantum.com/hubfs/White%20Papers/Graphics/2024-08-08%20A%20Future-Proof%20Cybersecurity%20Posture/2024-08-08%2015%20Webinar_%20A%20Future-Proof%20Cybersecurity%20Posture.png

Source: www.aliroquantum.com